ECEN 4/5632 Theory and Application of Digital Filters

Lecture 14, 10-27-26

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* With generous help from Gemini Pro and ChatGPT

WEEK 8 ● LECTURE 14

Upsampling, Interpolation, and Rational Rate Changes

From zero insertion to sample-rate conversion and transposed convolutions



Oppenheim & Schafer §4.6

ROADMAP

Learning objectives

The goal is to design complete sample-rate conversion chains and connect them to learned upsampling.

- Construct an upsampled sequence by zero insertion.
- Explain why the DTFT becomes X(e^{jωL}) and why spectral images appear.
- Choose and scale an interpolation lowpass filter.
- Build an L/M rational rate converter and select its cutoff.
- Relate ConvTranspose1d to zero insertion + convolution and diagnose periodic artifacts.

Pacing

0–20 min: upsampling
20–45 min: interpolation
45–60 min: L/M conversion
60–75 min: transposed conv

Core question

After inserting zeros, what must the filter do in both frequency and amplitude?

01

Upsampling by zero insertion

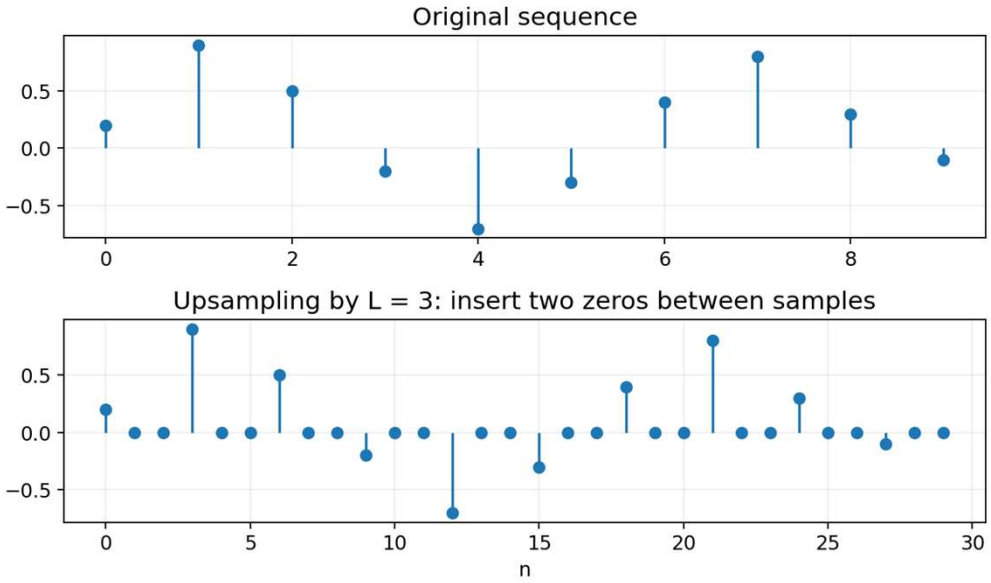
Create a higher-rate sequence by inserting L−1 zeros between samples.

0:02–0:08

Upsampling: the time-domain definition

$$x_i[n] = \begin{cases} x[n/L], & n = 0, \pm L, \pm 2L, \dots \\ 0, & \text{otherwise} \end{cases}$$

- Output sample rate becomes L F_s.
- Original samples remain at indices n = kL.
- Inserted zeros are placeholders—not estimates of the missing waveform.
- Interpolation filtering will fill in meaningful intermediate values.

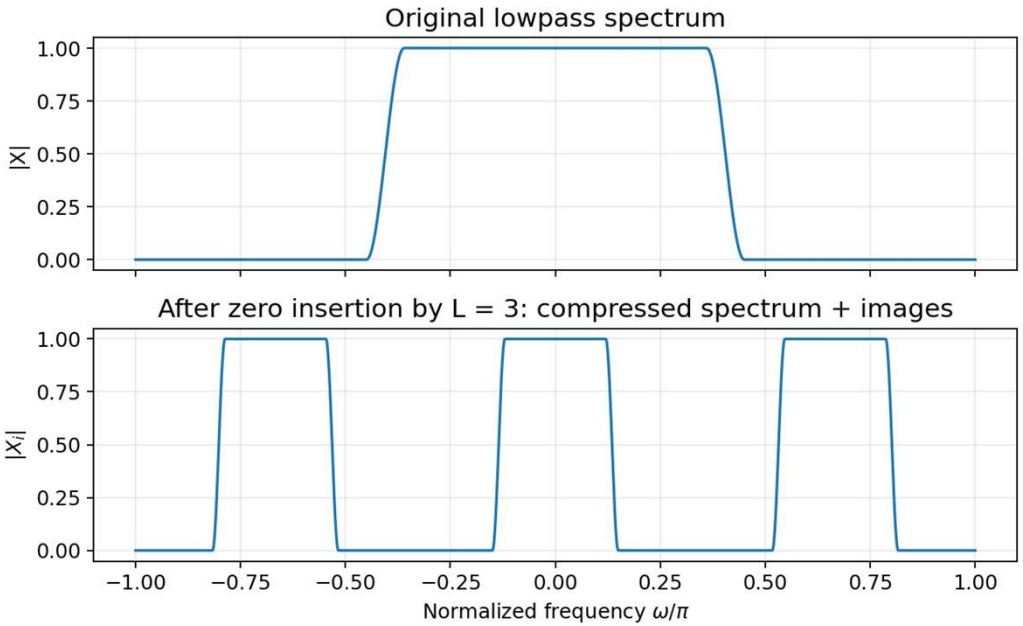


0:08–0:15

Frequency-domain effect: compression + images

$$X_i(e^{j\omega}) = X(e^{j\omega L})$$

- **Compression:** the original spectrum is squeezed by L.
- **Periodicity:** the compressed copy repeats L times over [−π, π].
- **Images:** the extra replicas are artifacts of zero insertion.
- **Goal:** keep the baseband copy; remove the rest.



CONCEPT CHECK

Concept check: image locations

If L = 3 and the original signal is baseband, how many compressed spectral copies appear over one 2π interval after zero insertion?

Reveal after discussion

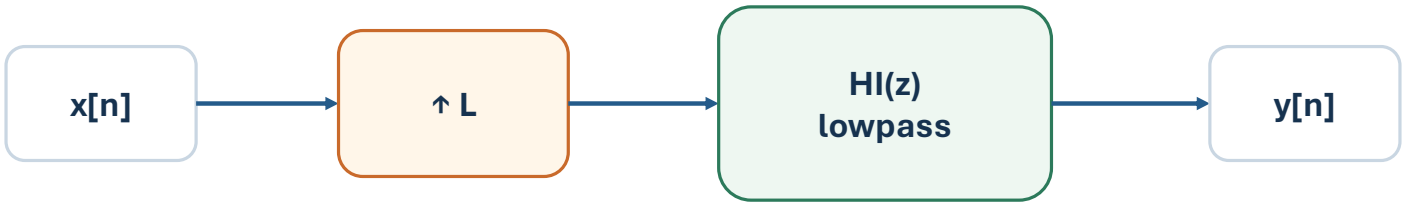
02

Interpolation = upsample + lowpass

The anti-imaging filter isolates the desired baseband replica and reconstructs intermediate samples.

0:20–0:28

The interpolation filter



Ideal cutoff

$\omega_c = \pi/L$

Pass only one image

The lowpass retains the compressed baseband copy centered at $\omega = 0$.

DC gain

Set $H_I(e^{j0}) = L$ so the reconstructed sequence preserves amplitude.

0:28–0:34

Why the gain must be L

Test signal: x[n] = 1 for all n.

After ↑L

Sequence is 1, 0, 0, ..., 1, 0, 0, ...
Average value = 1/L

→

Lowpass with unit DC gain

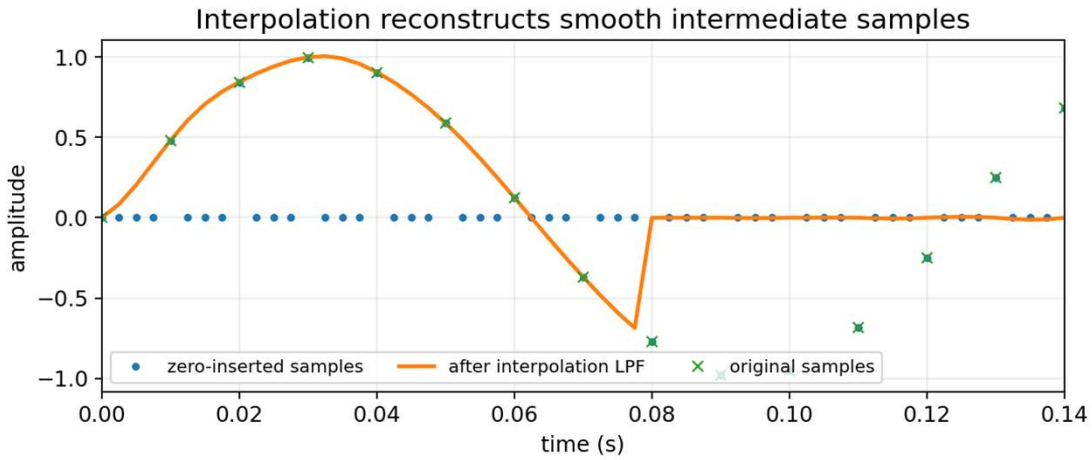
Smoothing preserves the reduced average → output tends toward 1/L.

Correct interpolation filter

Multiply the lowpass by L so constant input stays constant: y[n] = 1.

0:34–0:40

Interpolation in the time domain



- Inserted zeros disappear after filtering.
- Original sample locations are retained (subject to filter delay / edge effects).
- Intermediate samples are weighted combinations of neighboring originals.
- Ideal sinc interpolation is conceptually exact for bandlimited signals; FIR approximations are practical.

DSP interpretation

The filter is reconstructing a smoother sequence on a denser sampling grid.

0:40–0:45

Choosing an interpolation FIR

Passband

Preserve the compressed baseband: $|\omega| \le \omega_p < \pi/L$.

Stopband

Suppress the first image beginning near π/L.

Gain

Scale coefficients so the DC gain is L.

Delay

For N-tap linear-phase FIR, delay = $(N-1)/2$ high-rate samples.

Implementation check: feed a constant signal and verify the output amplitude is unchanged away from transients.

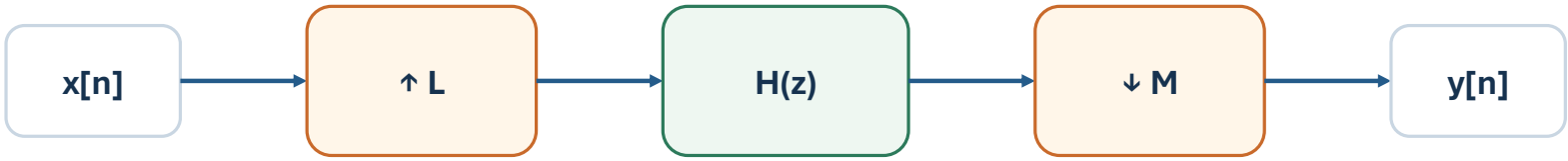
03

Rational sample-rate conversion L/M

Upsample first, filter once, then downsample.

0:45–0:51

Why the order is ↑L then ↓M



$$F_{s,out} = \frac{L}{M} F_{s,in}$$

- Upsampling first makes the intermediate grid fine enough to represent the desired output sample times.
- A single lowpass filter can remove interpolation images and prevent later decimation aliasing.
- Reduce L/M to lowest terms before implementation.

0:51–0:56

One filter serves two jobs

$$\omega_c = \min\left(\frac{\pi}{L}, \frac{\pi}{M}\right)$$

Anti-imaging constraint

After ↑L, reject replicas outside the desired compressed baseband.

Anti-aliasing constraint

Before ↓M, remove content that lies above the new Nyquist limit.

Combined design

Use the stricter cutoff. In practice, reserve a transition band below the ideal boundary.

0:56–1:00

Real example: 48 kHz → 44.1 kHz

Rate ratio

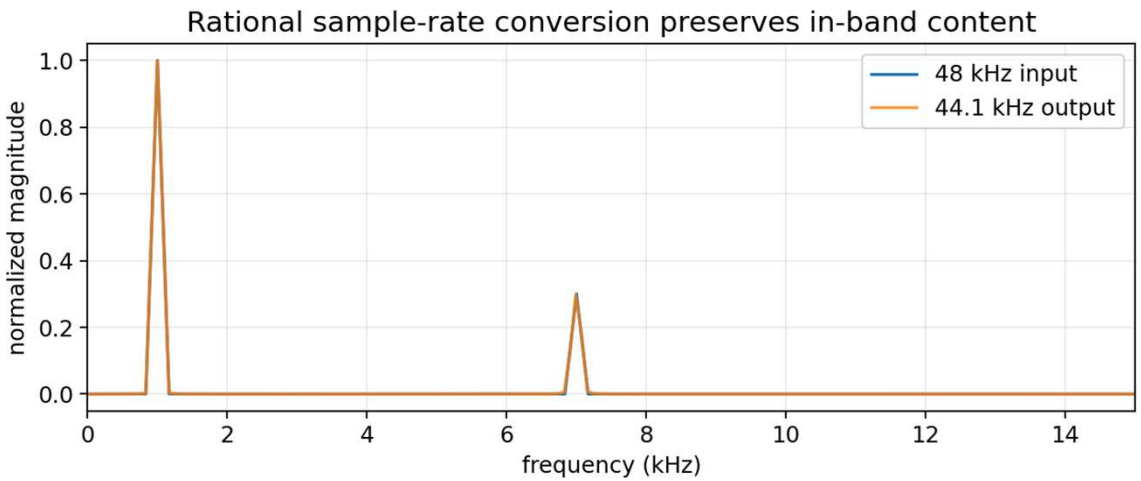
44,100 / 48,000 = 147 / 160

Conceptual chain

↑147 → lowpass → ↓160

Efficient implementation

Polyphase filtering computes only the output samples that are needed.



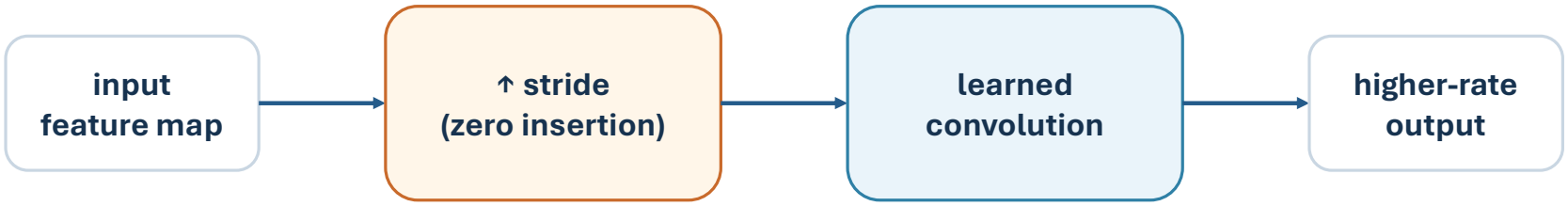
04

Transposed convolutions and learned upsampling

A stride in the reverse direction is closely related to zero insertion followed by filtering.

1:00–1:06

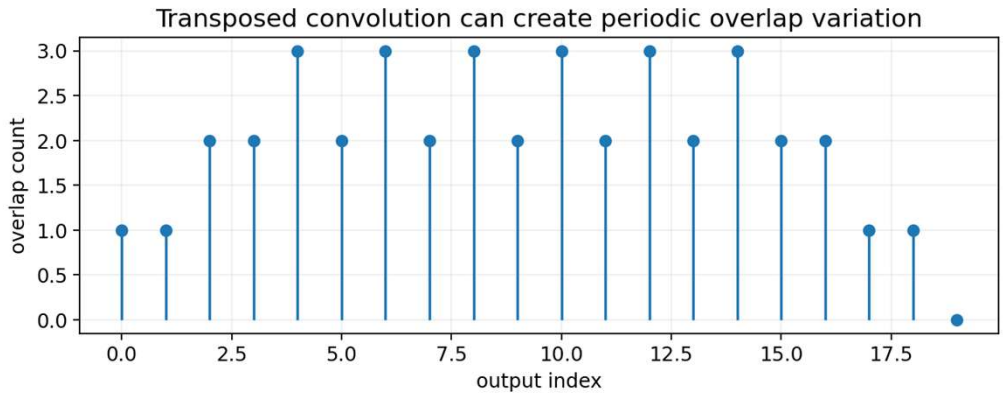
ConvTranspose1d: a DSP interpretation



- For stride > 1, transposed convolution can be viewed as inserting zeros and then applying a learned filter (with indexing/padding details).
- The learned filter is not automatically an anti-imaging filter.
- Uneven overlap or poorly controlled imaging can create periodic artifacts.

1:06–1:11

Where periodic / checkerboard artifacts come from



- **Uneven overlap:** some output positions receive more kernel contributions than others.
- **Spectral images:** zero insertion creates replicas that a learned kernel may not suppress.
- **Training can exploit phase patterns:** periodic structure may become visible or audible.

Mitigation patterns

Resize/upsample + ordinary convolution, kernel/stride choices with even overlap, or explicit lowpass / anti-imaging constraints.

1:11–1:14

A DSP-informed initialization idea

Start from interpolation

Initialize each upsampling kernel with a smooth lowpass / linear-interpolation-like response.

Then learn

Allow gradient descent to adapt the kernel while regularization or architecture preserves reasonable spectral behavior.

Why it helps

- The model begins with less imaging energy.
- Early training outputs can be smoother and more stable.
- The initialization encodes a physically sensible prior without preventing learning.

1:14–1:15

Lecture 14 wrap-up

- **Upsampling** inserts zeros and creates spectral images.
- **Interpolation** lowpass filters those images and uses gain L.
- **Rational conversion** uses ↑L → LPF → ↓M with cutoff min(π/L, π/M).
- **ConvTranspose1d** is closely related to zero insertion + learned filtering.
- **DSP insight** helps explain and mitigate periodic artifacts in learned upsampling.

Exit question

Why does an interpolation filter need gain L rather than unity gain?

Weekly work

Homework + quiz solutions are in the instructor packet.